—National Institute of Standards and Technology

Also, NIST defines the following five essential characteristics for cloud computing:

- *On-demand self-service*—The cloud enables us to provision resources ad hoc with the click of a button or an API call.
- *Broad network access*—Capabilities are available over the network.
- *Resource pooling*—The cloud assigns resources based on a multitenant model, which means consumers share the same physical and virtual resources.
- *Rapid elasticity*—The cloud allows us to expand and shrink the provisioned capacity on demand.
- *Measured service*—The cloud provides metrics allowing consumers to gain insights into the utilization of their resources.

Besides that, offerings are often divided into the following three types:

- *Public*—A cloud managed by an organization and open to use by the general public
- *Private*—A cloud that virtualizes and distributes the IT infrastructure for a single organization
- *Hybrid*—A mixture of a public and a private cloud

Amazon Web Services (AWS) is a public cloud. By combining your on-premises data center with AWS, you are building a hybrid cloud.

Cloud computing services also have several classifications, described here: